# Visualization Literacy Analysis

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## Read in data files

```r
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)


# fast_RTs <- data.frame(ParticipantId = character(),
#                        TrialName = character(),
#                        type = character(),
#                        time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
     select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
    rename(Correct_1 = correct, Correct_2 = "...9") %>%
    mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)
```

```r
  all_data <- all_data %>%
    bind_rows(new_temp)

}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type",TrialType))
```

## Basic checks

```r
#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  kable()
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

| Condition | nsubs |
|---|---|
| VR | 50 |
| VR Monitor | 39 |
| VR Monitor Stereo | 33 |

```r
#why is the balance so off?
```

## Remove outliers based on RT

```r
#removing on trial-by-trial basis


#remove answerRTs below 2000ms first
all_data_remove <- all_data %>%
  filter(AnswerRT >= 2000)
dim(all_data)[1]
```

```
## [1] 2074
```

```r
dim(all_data_remove)[1]
```

```
## [1] 2067
```

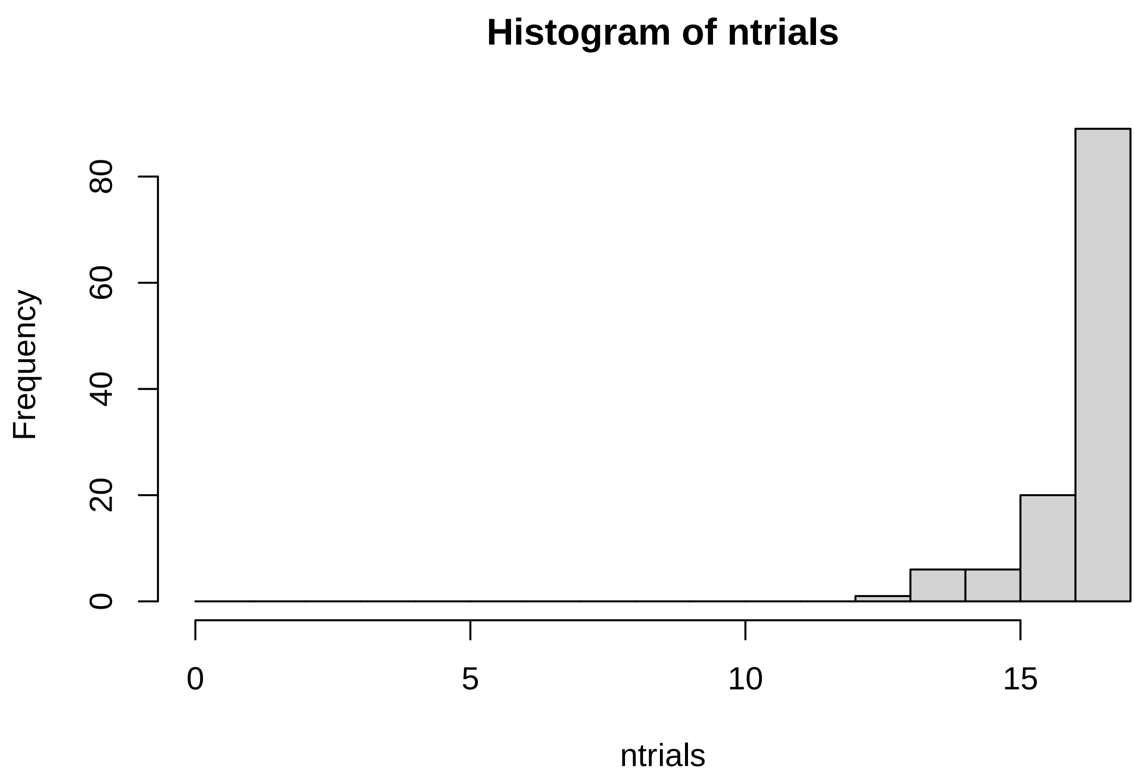
```r
#drops 7 trials

rt_data_summary <- all_data %>%
  group_by(TrialName) %>%
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),
            sdAnswerRT = sd(AnswerRT, na.rm = T),
            UB = meanAnswerRT + 3*sdAnswerRT,
            LB = meanAnswerRT - 3*sdAnswerRT)
rt_data_summary
```

```
## # A tibble: 17 x 5
##    TrialName     meanAnswerRT sdAnswerRT      UB      LB
##    <chr>                <dbl>      <dbl>   <dbl>   <dbl>
##  1 BarChartQ1          27611.     14762.  71897. -16675.
##  2 BarChartQ2          16921.     13338.  56935. -23093.
##  3 BarChartQ3          15609.      9948.  45452. -14235.
##  4 BarChartQ4          10288.      8978.  37222. -16646.
##  5 LineChartQ1         49185.     29505. 137699. -39329.
##  6 LineChartQ2         40127.     31660. 135107. -54854.
##  7 LineChartQ3         27190.     17504.  79701. -25322.
##  8 LineChartQ4         14779.     16568.  64483. -34925.
##  9 LineChartQ5         27877.     17250.  79628. -23874.
## 10 ScatterplotQ1       32801.     24294. 105682. -40080.
## 11 ScatterplotQ2       29636.     23980. 101576. -42304.
## 12 ScatterplotQ3       45580.     33014. 144623. -53463.
## 13 ScatterplotQ4       35987.     25402. 112194. -40219.
## 14 ScatterplotQ5       59805.     42684. 187859. -68248.
## 15 SurfacePlotQ1       56608.     36042. 164733. -51516.
## 16 SurfacePlotQ2       65110.     50062. 215297. -85076.
## 17 SurfacePlotQ3       48373.     37424. 160646. -63900.
```

```r
all_data_no_outliers <- all_data_remove %>%
  group_by(TrialName) %>%
  filter((!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT))))
dim(all_data_no_outliers)[1]
```

```
## [1] 2020
```

```r
#drops 47 more trials

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:17))
```

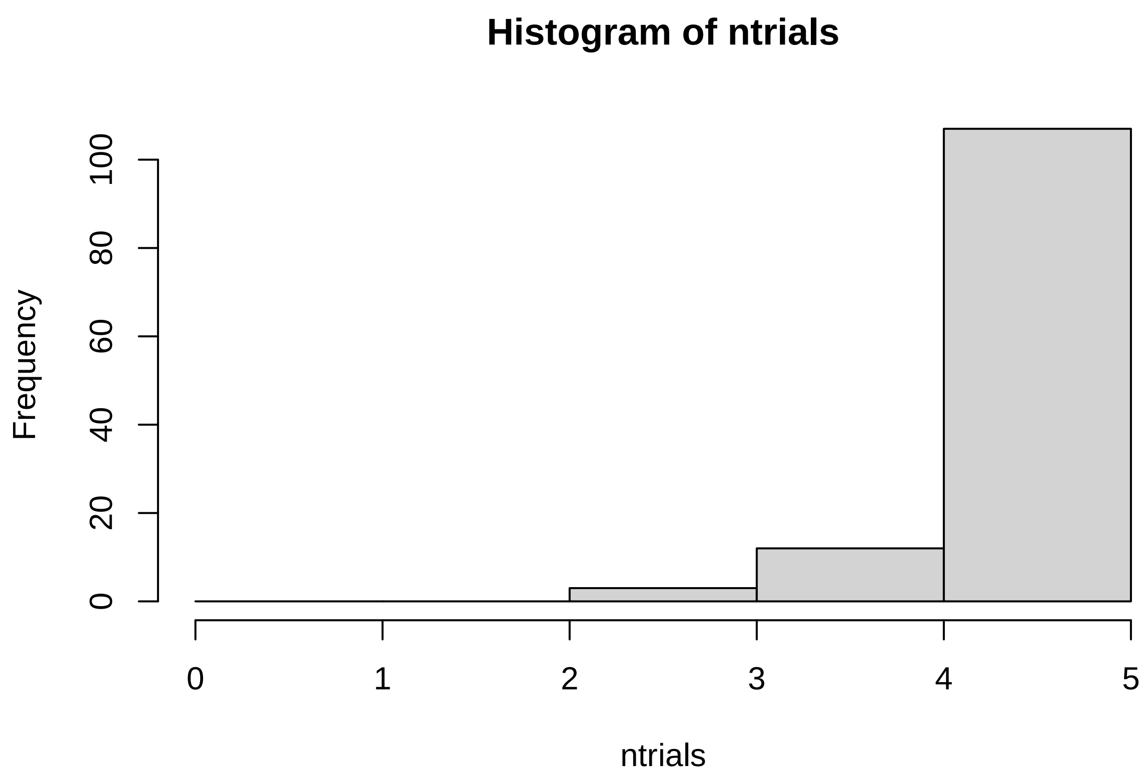



```
##maybe consider replacing outliers with means instead of removing them?
```

### Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

```
all_data_no_outliers <- all_data_no_outliers %>%
  filter(grepl("ScatterplotQ[0-9]", TrialName))

all_data_no_outliers %>%
  group_by(ParticipantId) %>%
  summarize(ntrials = n()) %>%
  with(hist(ntrials, breaks = 0:5))
```




### Compare RTs for conditions and question type (three types: identify, relate, predict)

```
#compare read times (should be no differences of condition)
#compare answer times (potentially a difference)


trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_readRT = mean(readRT),
            mean_answerRT = mean(AnswerRT),
            n = n())
```

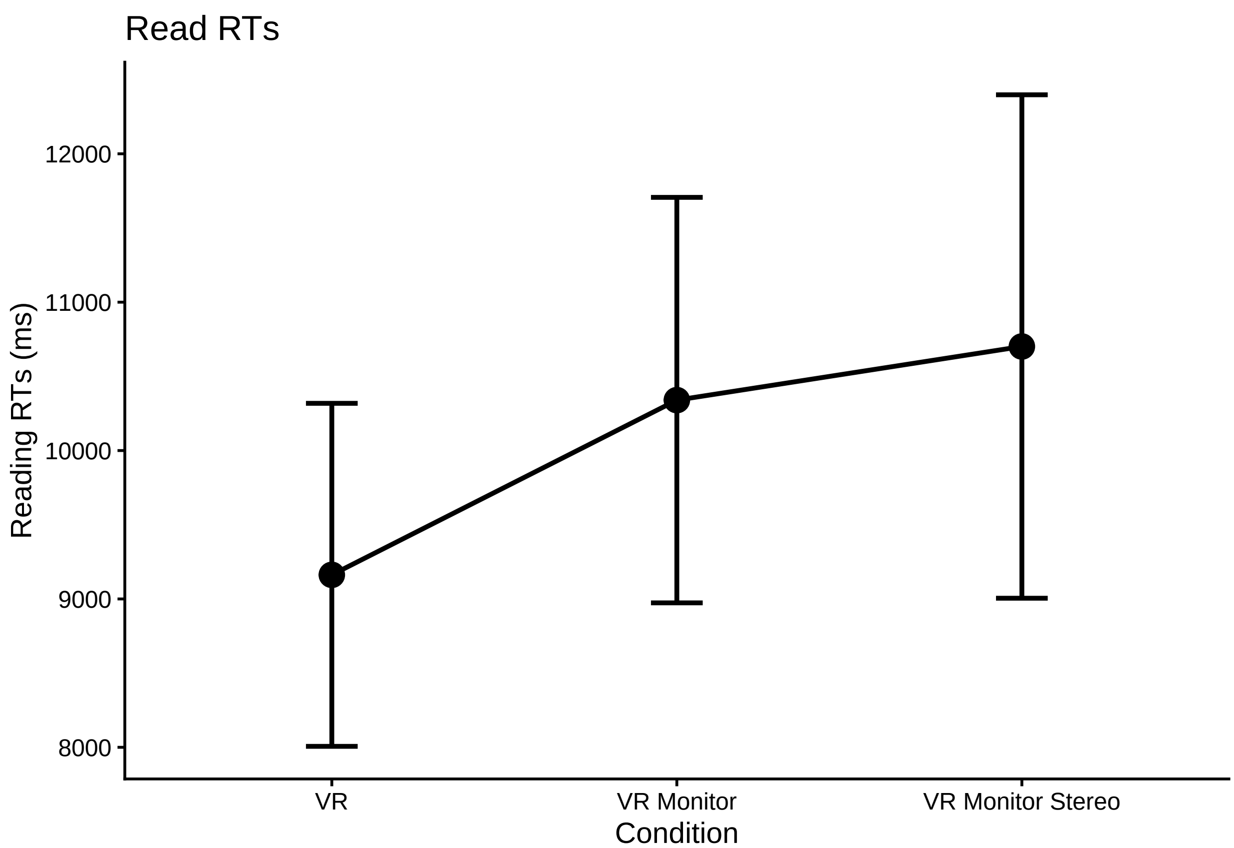
```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
# first, make .csv files in wide format to double check in statview
read_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_readRT) %>%
  pivot_wider(names_from = TrialType,values_from = mean_readRT)

# fill in na values with mean for type1, type2 and type3 RT
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T
read_rt_type_wider_clean <- read_rt_type_wider %>% replace_na(list(Type1 = mean(read_rt_type_wider$Type
read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type2 = mean(read_rt_type_wide
```

```
read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type3 = mean(read_rt_type_wide

answer_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType,values_from = mean_answerRT)
write.csv(read_rt_type_wider, file = "readtypeRTs.csv", row.names = F)
write.csv(answer_rt_type_wider, file = "answertypeRTs.csv", row.names = F)

# fill in na values with mean for type1, type2 and type3 RT
answer_rt_type_wider_clean <- answer_rt_type_wider %>% replace_na(list(Type1 = mean(answer_rt_type_wide
answer_rt_type_wider_clean <- answer_rt_type_wider_clean %>% replace_na(list(Type2 = mean(answer_rt_type
answer_rt_type_wider_clean <- answer_rt_type_wider_clean %>% replace_na(list(Type3 = mean(answer_rt_type


#### READ RTs ####

#make some plots

#just condition main effect
readplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(readRT)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

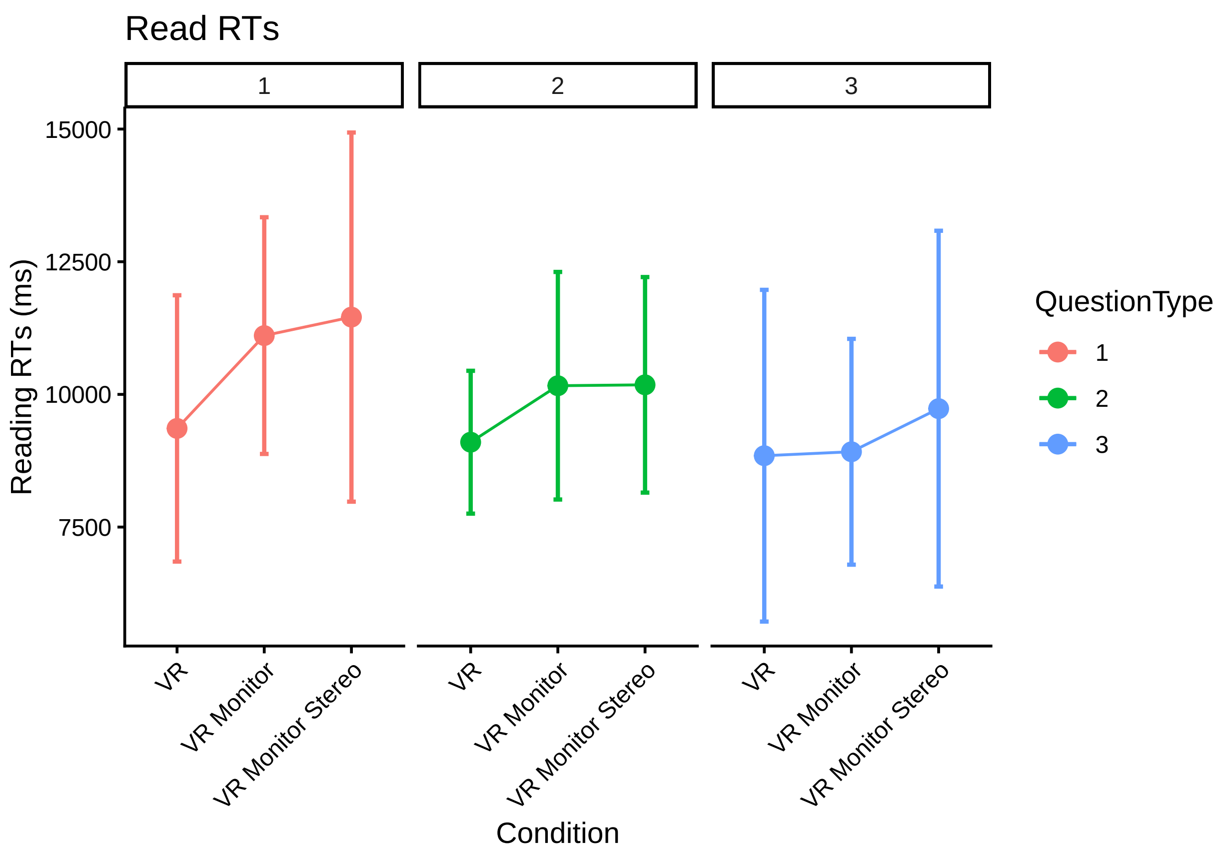
```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```
 ggplot(readplot1, aes(Condition,
                       overall_condition_mean,
                       group = 1,
                       ymin = overall_condition_mean - CI,
                       ymax = overall_condition_mean + CI)) +
   theme_classic() +
   geom_point(size = 4) +
   geom_errorbar(width = .15, linewidth = 0.85) +
   geom_line(linewidth = 0.85) +
   labs(y = "Reading RTs (ms)", title = "Read RTs")
```




```
#make a little plot using wide format
superbPlot(read_rt_type_wider_clean,
    BSFactors    = "Condition",
    WSFactors    = "QuestionType(3)",
    variables    = c("Type1", "Type2", "Type3"),
    statistic    = "meanNArm",
    errorbar     = "CI",
    gamma        = 0.95,
    adjustments = list(
        purpose       = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Reading RTs (ms)", title = "Read RTs")
```

```
## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".
```




```r
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
     select(ParticipantId, Condition, TrialType, mean_readRT) %>%
     pivot_wider(names_from = TrialType, values_from = mean_readRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
readRT_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_readRT")

read_rt_type_anova <- aov_ez(id = "ParticipantId",
                             dv = "mean_readRT",
                             data = readRT_means_clean,
                              within = "TrialType",
                              between = "Condition",
                             anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```r
kable(nice(read_rt_type_anova))
```

| Effect | df | MSE | F | pes | p.value |
|---|---|---|---|---|---|
| Condition | 2, 119 | 56889167.71 | 1.09 | .018 | .340 |
| TrialType | 1.59, 188.79 | 24878890.94 | 3.29 + | .027 | .051 |
| Condition:TrialType | 3.17, 188.79 | 24878890.94 | 0.47 | .008 | .711 |

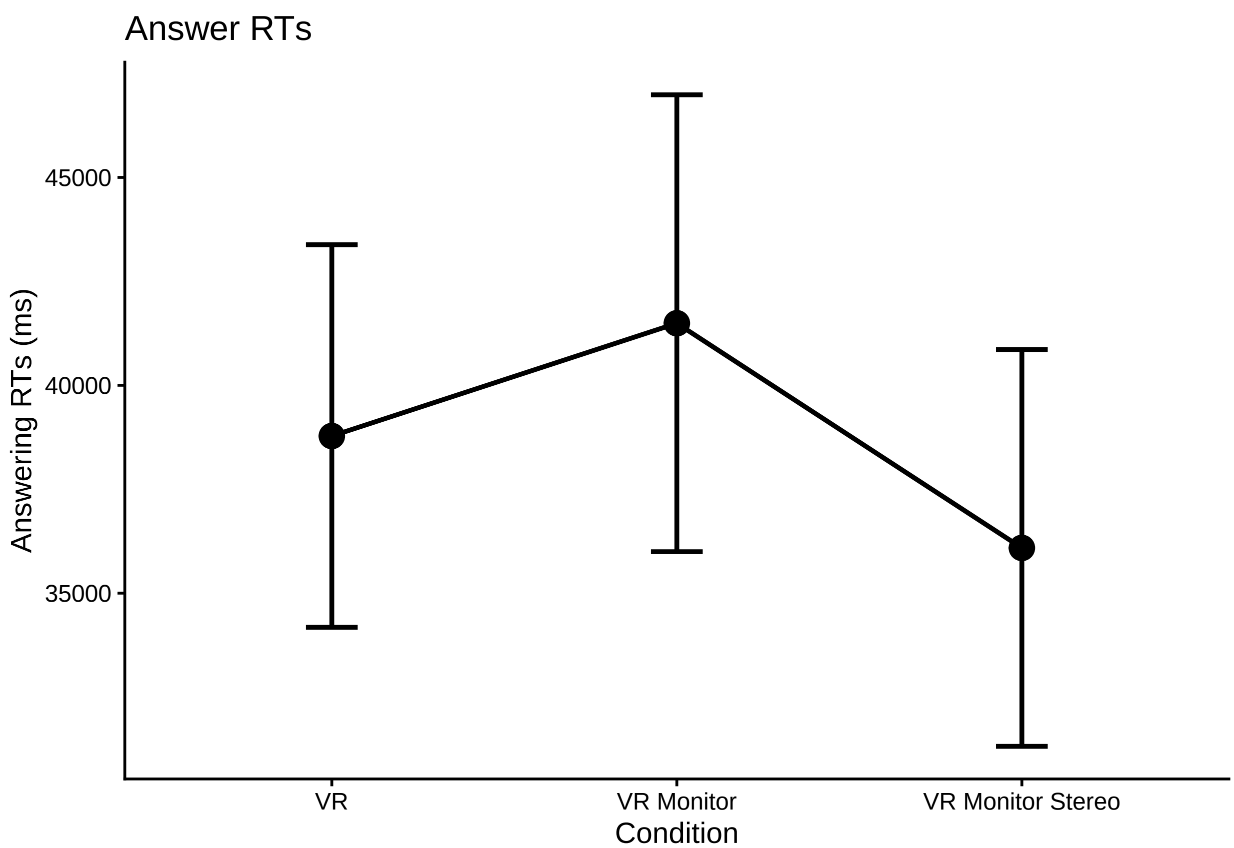
```r
#posthoc test for trial type
pairs(emmeans(read_rt_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate  SE  df t.ratio p.value
##  Type1 - Type2      827 456 119   1.815  0.1691
##  Type1 - Type3     1476 704 119   2.098  0.0946
##  Type2 - Type3      649 544 119   1.192  0.4600
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

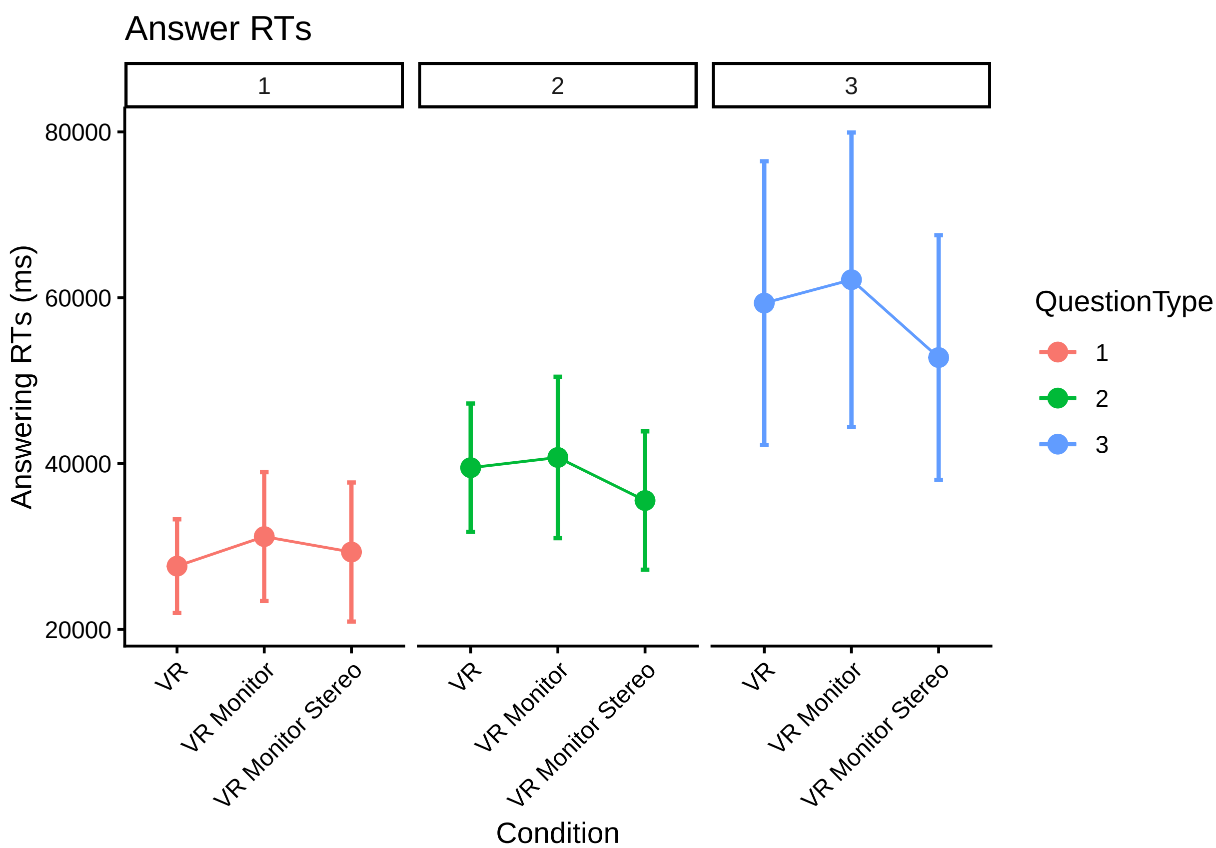
```r
#### ANSWER RTs ####

#make some plots

#just condition main effect
answerplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(AnswerRT)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
             n = n(),
             CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```r
 ggplot(answerplot1, aes(Condition,
                       overall_condition_mean,
                       group = 1,
                       ymin = overall_condition_mean - CI,
                       ymax = overall_condition_mean + CI)) +
   theme_classic() +
   geom_point(size = 4) +
   geom_errorbar(width = .15, linewidth = 0.85) +
   geom_line(linewidth = 0.85) +
   labs(y = "Answering RTs (ms)", title = "Answer RTs")
```




```r
#make the little plot
superbPlot(answer_rt_type_wider_clean,
    BSFactors    = "Condition",
    WSFactors    = "QuestionType(3)",
    variables    = c("Type1", "Type2", "Type3"),
    statistic    = "meanNArm",
    errorbar     = "CI",
    gamma        = 0.95,
    adjustments = list(
        purpose        = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Answering RTs (ms)", title = "Answer RTs")
```




```
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
     select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
     pivot_wider(names_from = TrialType, values_from = mean_answerRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
answerRT_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_answerRT")

answer_rt_type_anova <- aov_ez(id = "ParticipantId",
                               dv = "mean_answerRT",
                               data = answerRT_means_clean,
                                within = "TrialType",
                                between = "Condition",
                               anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```
kable(nice(answer_rt_type_anova))
```

| Effect | df | MSE | F | pes | p.value |
|---|---|---|---|---|---|
| Condition | 2, 119 | 972114036.18 | 0.83 | .014 | .439 |
| TrialType | 1.47, 175.41 | 749109866.00 | 46.16 *** | .279 | <.001 |
| Condition:TrialType | 2.95, 175.41 | 749109866.00 | 0.38 | .006 | .767 |

```
#posthoc test for trial type
pairs(emmeans(answer_rt_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate   SE  df t.ratio p.value
##  Type1 - Type2    -9209 1940 119  -4.737  <.0001
##  Type1 - Type3   -28718 3540 119  -8.104  <.0001
##  Type2 - Type3   -19510 3410 119  -5.724  <.0001
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)

ref <- emmeans(answer_rt_type_anova,~Condition|TrialType)

pairs(ref, adjust = "Tukey")
```

```
## TrialType = Type1:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                    -3564 3370 119  -1.059  0.5416
##  VR - VR Monitor Stereo             -1703 3530 119  -0.482  0.8800
##  VR Monitor - VR Monitor Stereo      1861 3730 119   0.499  0.8718
##
## TrialType = Type2:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                    -1236 4120 119  -0.300  0.9515
##  VR - VR Monitor Stereo              3960 4320 119   0.916  0.6310
##  VR Monitor - VR Monitor Stereo      5196 4560 119   1.140  0.4913
##
## TrialType = Type3:
##  contrast                        estimate   SE  df t.ratio p.value
##  VR - VR Monitor                    -2816 8150 119  -0.345  0.9364
##  VR - VR Monitor Stereo              6570 8560 119   0.767  0.7236
##  VR Monitor - VR Monitor Stereo      9386 9030 119   1.040  0.5534
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#plot and interaction suggests that the conditions have different effects for different
#question types. posthoc tests indicate a difference between VR and VRMonitor for QType 1
```

### Accuracy

```
# do interrater reliability: Fits each trial as independent with two raters
# That is, this ignores participants (complete pooling, which may be problematic)
# 1410 trials from N = 85 participants, not exactly 17 trials per participant
# b/c of excluded trials
  irr::icc(select(ungroup(all_data_no_outliers), Correct_1, Correct_2))
```

```
##  Single Score Intraclass Correlation
```

```
##
##    Model: oneway
##    Type : consistency
##
##    Subjects = 592
##      Raters = 2
##      ICC(1) = 0.954
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(591,592) = 42.1 , p = 5.41e-311
##
##  95%-Confidence Interval for ICC Population Values:
##   0.946 < ICC < 0.96
```

```
  head(all_data_no_outliers)
```

```
## # A tibble: 6 x 11
## # Groups:   TrialName [5]
##   ParticipantId Condition  TrialNumber TrialName     readRT totalRT UserInput    Correct_1
##   <chr>         <chr>            <dbl> <chr>          <dbl>   <dbl> <chr>            <dbl>
## 1 AA0176        VR Monitor           6 ScatterplotQ1 12366.  27154. High income          1
## 2 AA0176        VR Monitor           7 ScatterplotQ2 11326.  23702. Low income           0
## 3 AA0176        VR Monitor           8 ScatterplotQ3  9830.  93848. Positive             0
## 4 AA0176        VR Monitor           9 ScatterplotQ4 19927.  94318. Decreasing           1
## 5 AA0176        VR Monitor          10 ScatterplotQ5  8245.  86052. Try cutting~         1
## 6 AA0876        VR                  13 ScatterplotQ1  2435.  40378. 2016.0               0
## # i 3 more variables: Correct_2 <dbl>, AnswerRT <dbl>, TrialType <chr>
```

```
# account for non-independence?
  #Data has to be narrow
#https://cran.r-project.org/web/packages/cccrm/index.html
icc.dat.narrow <- all_data_no_outliers %>%
                    pivot_longer(cols = c('Correct_1', 'Correct_2'))

colnames(icc.dat.narrow)[10] <- "Rater"
icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)

#Longitudinal concordance for two raters
#By all 17 trials (by items)
# icc.rm.trial.name <- ccclon(icc.dat.narrow,
#                              "value", "ParticipantId", "TrialName", "Rater")
#  icc.rm.trial.name
#  summary(icc.rm.trial.name)

#By time using trial numbers (by time)
 icc.rm.trial.number <- ccclon(icc.dat.narrow,
                               "value", "ParticipantId", "TrialNumber", "Rater" )
```

```
## Warning: `ccclon()` was deprecated in cccrm 3.0.0.
## i Please use `ccc_vc()` instead.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```

```
 icc.rm.trial.number
```

```
## CCC estimated by variance components:
##         CCC   LL CI 95%   UL CI 95%      SE CCC
```

```
## 0.949385915 0.940356117 0.957078885 0.004248889
```

```r
summary(icc.rm.trial.number)
```

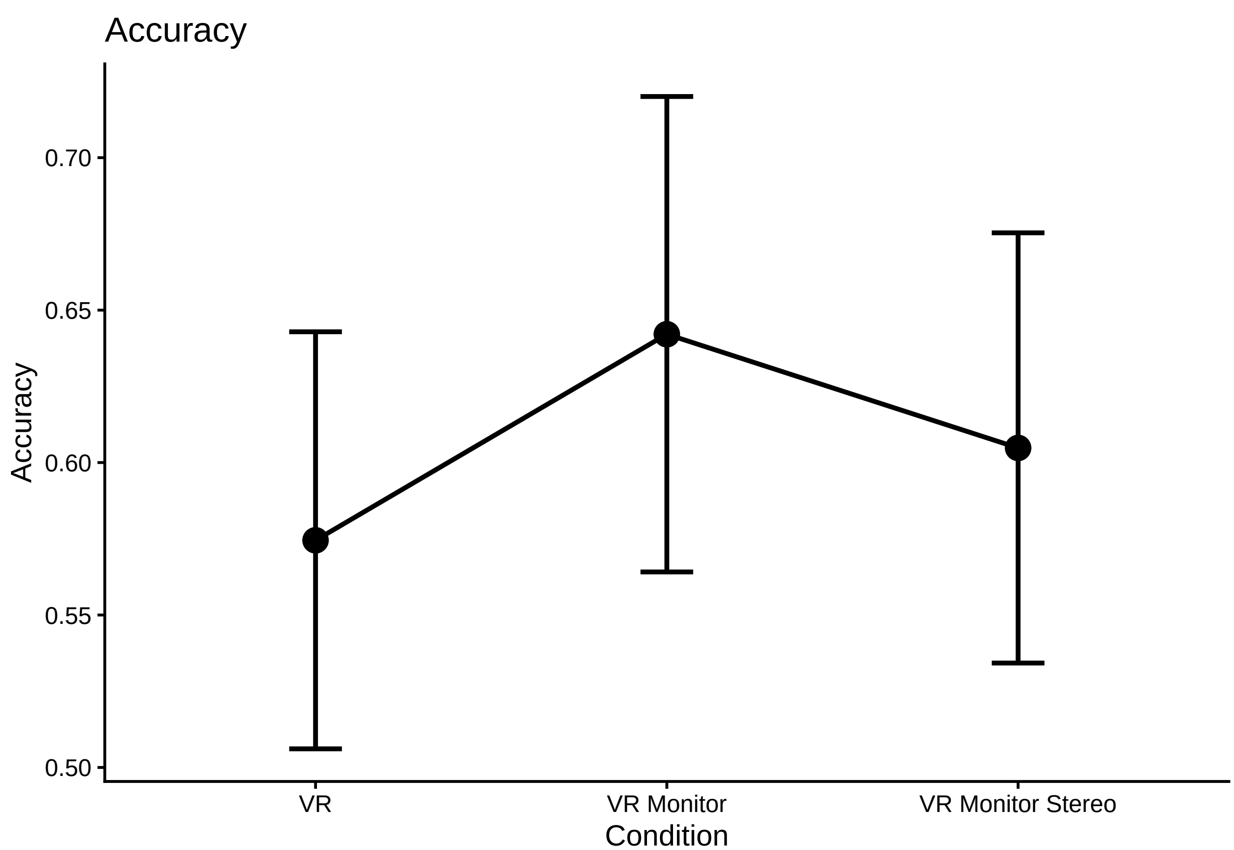
```
##        Subjects Subjects-Method   Subjects-Time          Method           Error
##    0.0127425554    0.0009092681    0.1821563081    0.0004085254    0.0090727415
##
## CCC estimated by variance components
##         CCC    LL CI 95%   UL CI 95%      SE CCC
## 0.949385915 0.940356117 0.957078885 0.004248889
```

```r
# A similar approach
#https://peerj.com/articles/9850/
# test.mod <-
#  lcc(data = hue, subject = "Fruit", resp = "H_mean",
#                   method = "Method", time = "Time", qf = 2, qr = 1)
# summary(test.mod)

# icc.dat.narrow$ParticipantId <- as.factor(icc.dat.narrow$ParticipantId)
# icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)
#
#
# icc.rm.another <-
# lcc(data = icc.dat.narrow,
#       subject = "ParticipantId",
#       resp = "value",
#       method = "Rater",
#       time = "TrialNumber",
#       qf = 1, #polynomial trends (1 to 3)
#       qr = 0, #random effects (0 is random int only, default)
#       components = TRUE)
#
# #LCC = Longitudinal Concordance Correlation
# #LPC = Longitudinal Pearson Correlation
# #LA  = Longitudinal Accuracy
# icc.rm.another
#
# #Model coefficients near 1 but only the the last one has a correctly
# #estimated model. Maybe because it's more or less flat over TrialNumber?
# lccPlot(icc.rm.another, type = "lpc")
# lccPlot(icc.rm.another, type = "lcc")
# lccPlot(icc.rm.another, type = "la")


all_data_no_outliers <- all_data_no_outliers %>%
  rowwise() %>%
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))

#just condition main effect
accplot1 <- all_data_no_outliers %>%
   group_by(ParticipantId, Condition) %>%
   summarize(overallmean = mean(Correct_Avg)) %>%
   group_by(Condition) %>%
   summarize(overall_condition_mean = mean(overallmean),
             se = std.error(overallmean),
```

```
            n = n(),
            CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```
ggplot(accplot1, aes(Condition,
                      overall_condition_mean,
                      group = 1,
                      ymin = overall_condition_mean - CI,
                      ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Accuracy", title = "Accuracy")
```

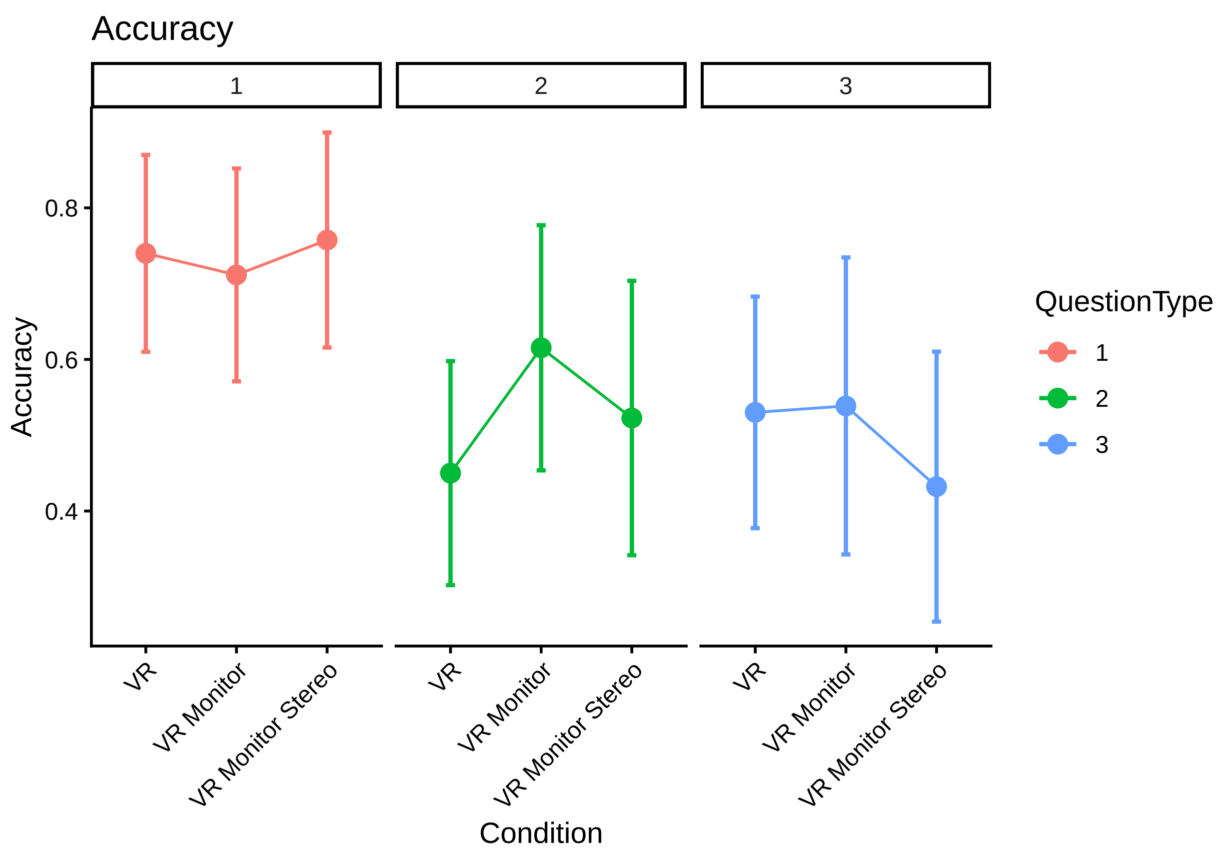



```
trial_type_mean_acc <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_acc = mean(Correct_Avg),
            n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
acc_type_wider <- trial_type_mean_acc %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
```

```r
  pivot_wider(names_from = TrialType,values_from = mean_acc)
write.csv(acc_type_wider, file = "typeacc.csv", row.names = F)

# replace na with mean
acc_type_wider_clean <- acc_type_wider %>% replace_na(list(Type1 = mean(acc_type_wider$Type1, na.rm=TRUE
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type2 = mean(acc_type_wider$Type2, na.r
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type3 = mean(acc_type_wider$Type3, na.r

#make the little plot
superbPlot(acc_type_wider_clean,
    BSFactors   = "Condition",
    WSFactors   = "QuestionType(3)",
    variables   = c("Type1", "Type2", "Type3"),
    statistic   = "meanNArm",
    errorbar    = "CI",
    gamma       = 0.95,
    adjustments = list(
        purpose       = "difference"
    ),
    plotLayout = "line",
    factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Accuracy", title = "Accuracy")
```




```
# there are missing values in anova for some participants because they don't
# do all trial types necessarily.  Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_mean_acc %>%
     select(ParticipantId, Condition, TrialType, mean_acc) %>%
     pivot_wider(names_from = TrialType, values_from = mean_acc)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
acc_means_clean <- x_clean %>%
     pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_acc")

acc_type_anova <- aov_ez(id = "ParticipantId",
                             dv = "mean_acc",
                             data = acc_means_clean,
                              within = "TrialType",
                              between = "Condition",
                             anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
```

```
kable(nice(acc_type_anova))
```

| Effect | df | MSE | F | pes | p.value |
|---|---|---|---|---|---|
| Condition | 2, 119 | 0.17 | 0.58 | .010 | .562 |
| TrialType | 1.98, 235.91 | 0.10 | 18.87 *** | .137 | <.001 |
| Condition:TrialType | 3.96, 235.91 | 0.10 | 1.67 | .027 | .159 |

```r
#posthoc test for trial type
pairs(emmeans(acc_type_anova, "TrialType"), adjust = "Tukey")
```

```
##  contrast      estimate     SE  df t.ratio p.value
##  Type1 - Type2   0.2070 0.0432 119   4.788  <.0001
##  Type1 - Type3   0.2361 0.0399 119   5.910  <.0001
##  Type2 - Type3   0.0291 0.0425 119   0.684  0.7732
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```r
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)

ref <- emmeans(acc_type_anova,~Condition|TrialType)

pairs(ref, adjust = "Tukey")
```

```
## TrialType = Type1:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                   0.0285 0.0657 119   0.433  0.9018
##  VR - VR Monitor Stereo           -0.0176 0.0690 119  -0.255  0.9648
##  VR Monitor - VR Monitor Stereo  -0.0460 0.0727 119  -0.633  0.8022
##
## TrialType = Type2:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                  -0.1654 0.0772 119  -2.143  0.0856
##  VR - VR Monitor Stereo           -0.0727 0.0810 119  -0.898  0.6428
##  VR Monitor - VR Monitor Stereo   0.0927 0.0854 119   1.085  0.5255
##
## TrialType = Type3:
##  contrast                        estimate     SE  df t.ratio p.value
##  VR - VR Monitor                  -0.0085 0.0832 119  -0.102  0.9943
##  VR - VR Monitor Stereo            0.0979 0.0874 119   1.121  0.5032
##  VR Monitor - VR Monitor Stereo   0.1064 0.0921 119   1.155  0.4824
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```